Menstrual hygiene

Menstrual hygiene is vital for good health, well-being, dignity, empowerment and productivity of women. The impact of poor menstrual hygiene on girls is increased stress levels, fear and embarrassment during menstruation. This can keep girls inactive during such periods leading to absenteeism from school.

Clean and safe **absorbable clothing materials, sanitary napkins, pads, tampons and menstrual cups** have been identified as materials used to manage menstruation. Changing sanitary material 4-5 hours as per the requirement, provides comfort, cleanliness and protection from infections. It also helps in enhancing the quality of life of women during this period. Used sanitary napkins should be wrapped in paper and disposed. It should not be thrown in open areas or drain pipe of toilets. Flushing of sanitary napkins in the drain pipes causes choking of the drainage line leading to water pollution.

Disposal of Napkins

The ecofriendly way to dispose menstrual waste scientifically and hygienically is to destroy the sanitary napkins using incinerators. Measures are being taken to install incinerators and napkin vending machines in washrooms of schools , colleges and public facilities.

Menopause

Menopause is the phase in a women's life when ovulation and menstruation stops. The average age of menopause is 45-50 years. It indicates the permanent cessation of the primary functions of the ovaries.

2.4 Fertilization and implantation

Fertilization occurs when a haploid sperm fuses with a haploid ovum to form a fertilized egg or diploid zygote.

The sperms deposited in the female reproductive tract undergo **capacitation**, which is a biochemical event that enables the sperm to penetrate and fertilise the egg. Fertilization occurs only if the ovum and sperms are transported simultaneously to the ampullary isthmic junction of the fallopian tube.

Before a sperm can enter the egg, it must penetrate the multiple layers of granulosa (follicular) cells which are around the ovum forming the **corona radiata (Fig. 2.10).** The follicular cells are held together by an adhesive cementing substance called hyaluronic acid. The acrosomal membrane disintegrates releasing the proteolytic enzyme, **hyaluronidase** during sperm entry through the corona radiata and zona pellucida. This is called **acrosomal reaction**. Once Fertilization is accomplished, cortical granules from the cytoplasm of the ovum form a barrier called the **Fertilization membrane** around the ovum preventing further penetration of other sperms. Thus **polyspermy** is prevented.

The first cleavage produces two identical cells called **blastomeres**. These produce 4 cells, then 8 and so on. After 72 hours of Fertilization, a loose collection of cells forms a berry shaped cluster of 16 or more cells called the **morula (Fig. 2.11)**.

Under the influence of progesterone, smooth muscles of the fallopian tube relax and the dividing embryo takes 4-5 days to move through the fallopian tube into the uterine cavity and finally gets implanted in the uterine wall. At this point the embryo consists of a fluid filled hollow ball of about 100 cells, called the **blastocyst**. The blastocyst is composed of a single layer of large flattened cells called **trophoblast** and a small cluster of 20-30 rounded cells called the **inner cell mass**. The inner cell mass of the blastocyst develops into the **embryo** and becomes embedded in the endometrium of the uterus.

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